



Latent class analysis in PLS-SEM: A review and recommendations for future applications

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ABSTRACT

With the increasing prominence of partial least squares structural equation modeling (PLS-SEM) in business research, the use of latent class analyses for identifying and treating unobserved heterogeneity has also gained momentum. Researchers have introduced various latent class approaches in a PLS-SEM context, of which finite mixture PLS (FIMIX-PLS) plays a central role due to its ability to identify heterogeneity and indicate a suitable number of segments to extract from the data. However, applying FIMIX-PLS requires researchers to make several choices that, if incorrect, could lead to wrong results and false conclusions. Addressing this concern, we present the results of a systematic review of FIMIX-PLS applications published in major business research journals. Our review provides an overview of the interdependencies between researchers' choices and identifies potential problem areas. Based on our results, we offer concrete guidance on how to prevent common pitfalls when using FIMIX-PLS, and identify future research areas.

1. Introduction

In recent years, partial least squares structural equation modeling (PLS-SEM; Lohmöller, 1989; Wold, 1982) has become a standard method for analyzing the complex inter-relationships between observed and latent variables. PLS is a composite-based approach to SEM that emphasizes prediction when estimating complex models whose structures are designed to provide causal explanations (Hair, Risher, Sarstedt, & Ringle, 2019). As a limited information approach, PLS-SEM estimates solutions when the sample size is relatively small compared to that which many other statistical methods require (Hair, Sarstedt, & Ringle, 2019). The method also offers researchers a great deal of flexibility with model specification, for example, when estimating conditional process models (Sarstedt, Hair, Nitzl, Ringle, & Howard, 2020), higher-order constructs (Sarstedt, Hair, Cheah, Becker, & Ringle, 2019), or formatively specified measurement models (Sarstedt, Hair, Ringle, Thiele, & Gudergan, 2016). Finally, PLS-SEM supports using both in-sample and out-of-sample prediction metrics (Hair, 2021), whose joint consideration is crucial, given that a model performing well in terms of in-sample prediction does not necessarily have high levels of out-of-sample

predictive power (Sarstedt & Danks, 2021; Shmueli, 2010).

Numerous PLS-SEM reviews show that the method was widely disseminated during the last decade, especially in the social sciences (e. g., Ali, Rasoolimanesh, Sarstedt, Ringle, & Ryu, 2018; Ringle, Sarstedt, Mitchell, & Gudergan, 2020; Willaby, Costa, Burns, MacCann, & Roberts, 2015), but also in other fields of scientific inquiry, such as agricultural science, engineering, environmental science, and medicine (e. g., Avkiran, 2018; Durdyev, Ihtiyar, Banaitis, & Thurnell, 2018; Svensson et al., 2018; Menni et al., 2018; Sarstedt, 2019). These reviews also show that researchers applying PLS-SEM usually analyze the full data set, implicitly assuming that the data stem from a single homogeneous population. This assumption is often unrealistic, as individuals differ in their perceptions, attitudes, and behaviors (Jedidi, Jagpal, & DeSarbo, 1997). Failure to consider the data characteristics' heterogeneity could threaten PLS-SEM results' validity and lead to incorrect conclusions (Becker, Rai, Ringle, & Völckner, 2013; Sarstedt et al., 2020). For example, when researchers analyze data on the aggregate level, positive and negative group-specific effects might cancel one another, thereby triggering type II errors. While the source of such heterogeneity can sometimes be traced back to observable characteristics, heterogeneity

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is—more often than not—unobserved, since it does not depend on an a priori known observable characteristic or combinations of such characteristics (DeSarbo, Di Benedetto, Jedidi, & Song, 2006; Esposito Vinzi, Trinchera, Squillacioti, & Tenenhaus, 2008; Fordellone & Vichi, 2020).

To address this concern, PLS-SEM research offers a series of latent class techniques that facilitate the identification and treatment of unobserved heterogeneity. These techniques generalize, for example, finite mixture, genetic algorithm, and hill-climbing approaches to PLS-SEM—see Sarstedt et al. (2017) for an overview. Finite mixture partial least squares (FIMIX-PLS; Hahn, Johnson, Herrmann, & Huber, 2002) is the initial, and most widely applied, latent class approach. Drawing on the mixture regression concept, FIMIX-PLS simultaneously estimates group-specific path coefficients for a predefined number of segments and derives each observation's segment membership (Ringle, Wende, & Will, 2010; Sarstedt, Ringle, & Gudergan, 2016). Simulation studies show that FIMIX-PLS reveals the existence of heterogeneity in PLS path models reliably and correctly indicates the appropriate number of segments to retain from the data (Sarstedt, Becker, Ringle, & Schwaiger, 2011).

Nevertheless, FIMIX-PLS is clearly limited in terms of correctly identifying the underlying segment structure that the group-specific path coefficients define (Ringle, Sarstedt, & Schlittgen, 2014; Ringle, Sarstedt, Schlittgen, & Taylor, 2013), especially when the path model includes formatively specified measurement models (Becker, Rai, Ringle, & Völckner, 2013). Researchers have therefore called for the joint application of FIMIX-PLS and other latent class techniques, such as PLS prediction-oriented segmentation (PLS-POS; Becker, Rai, Ringle, & Völckner, 2013), since they excel in terms of segment-specific parameter recovery. As the only method offering researchers a clear indication of the number of segments to extract from the data, FIMIX-PLS plays a central role in this process (Sarstedt et al., 2017).

However, applying FIMIX-PLS requires researchers to make several choices that, if incorrect, could lead to wrong results. For example, research offers dozens of model selection criteria for determining the number of segments. However, these criteria differ vastly in terms of their ability to indicate the correct number of segments (Sarstedt, Becker, Ringle, & Schwaiger, 2011). In addition, researchers have

We identify several problem areas in researchers' FIMIX-PLS use regarding their (1) reporting and using of information and classification criteria to determine the appropriate number of segments, (2) hard-clustering of segments, (3) documenting and interpreting of segment-specific path coefficients, as well as (4) carrying out follow-up latent class and ex post analyses. Our findings suggest that researchers need to consider best practices in PLS-SEM-based latent class analyses to a greater extent, in order to ensure their results' validity and enhance their usefulness for business practice.

2. Finite mixture PLS (FIMIX-PLS)

Hahn, Johnson, Herrmann, and Huber (2002) originally introduced the FIMIX-PLS approach, which researchers extended in the following years (e.g., Rigdon, Ringle, & Sarstedt, 2010; Sarstedt, Becker, Ringle, & Schwaiger, 2011; Sarstedt, Ringle, & Hair, 2017). As its name implies, FIMIX-PLS follows a finite mixture approach by assuming that the data originate from several subpopulations (segments). Each segment is modeled separately and the overall population is a mixture of the segments (McLachlan & Peel, 2000). That is, each object is regarded as a realization of the mixture density with K segments, where

$$f_{ijk}(x_i|\theta_k) = \sum_{k=1}^K \rho_k f_{ijk}(x_i) \quad (1)$$

with ρ_k being the relative segment size ($\rho_k > 0 \forall k$, $\sum_{k=1}^K \rho_k = 1$), $f_{ijk}(\cdot)$ a density function, and θ_k the segment-specific vector of unknown parameters for segment k .

Following this concept, FIMIX-PLS relies on the following function form:

$$\eta_i = \sum_{k=1}^K \rho_k f_{ijk}(\eta_i|\xi_i, B_k, \Gamma_k, \Psi_k) \quad (2)$$

where η_i is a matrix of endogenous latent variable scores, B_k (Γ_k) is the path coefficient matrix of the endogenous (exogenous) latent variables, and Ψ_k is the matrix of each segment's regression variances in the structural model. Substituting $f_{ijk}(\eta_i|\xi_i, B_k, \Gamma_k, \Psi_k)$ results in:

$$\eta_i = \sum_{k=1}^K \rho_k \left[\frac{1}{(2\pi)^{Q/2} \sqrt{|\Psi_k|}} \right] \exp \left\{ -\frac{1}{2} ((I - B_k)\eta_i + (-\Gamma_k)\xi_i) \Psi_k^{-1} ((I - B_k)\eta_i + (-\Gamma_k)\xi_i) \right\} \quad (3)$$

emphasized that determining the number of segments requires considering multiple criteria, such as the segments' (relative) sizes and their degree of fuzziness (e.g., Hair, Sarstedt, Matthews, & Ringle, 2016).

Owing to the complexities of applying FIMIX-PLS, a systematic assessment of how prior research applied the technique could provide important guidance and—if necessary—opportunities for course correction in future applications. While there is no shortage of reviews of PLS-SEM use (e.g., Ali, Rasoolimanesh, Sarstedt, Ringle, & Ryu, 2018; Ghasemy, Teeroovengadam, Becker, & Ringle, 2020; Hair, Hollingsworth, Randolph, & Chong, 2017; Lin et al., 2020; Usakli & Kucukergin, 2018), there is as yet no such assessment of FIMIX-PLS. Our study fills this gap in research by systematically reviewing 45 FIMIX-PLS applications published until the end of 2020 in journals that Scopus ranks as belonging to the first two quartiles in business-related subject categories. The objective of this review is to provide an overview of the interdependencies of researchers' choices, in order to identify potential problem areas, and discuss their implications. Based on our results, we offer concrete guidance on how to prevent common pitfalls when using FIMIX-PLS and to identify areas for future research.

where Q depicts the number of endogenous latent variables in the structural model and I the identity matrix.

In FIMIX-PLS, parameter estimation relies on the expectation maximization algorithm (EM; Dempster, Laird, & Rubin, 1977), which maximizes the log-likelihood equation of (3), defined as:

$$LnL = \sum_{i=1}^N \sum_{k=1}^K z_{ik} \ln(f(\eta_i|\xi_i, B_k, \Gamma_k, \Psi_k)) + \sum_{i=1}^N \sum_{k=1}^K z_{ik} \ln(\rho_k) \quad (4)$$

where z_{ik} is 1 if object i belongs to segment k , 0 else. The relative segment size ρ_k , the parameters in ξ_i , B_k , Γ_k , and Ψ_k , as defined in Eq. (3), and the expected values $E(z_{ik}) = P_{ik}$ are computed according to Bayes' theorem—see Rigdon, Ringle, and Sarstedt (2010) for a detailed description of the EM algorithm in the context of FIMIX-PLS. The algorithm converges when there is no substantial improvement in the LnL (Eq. (4)) between two consecutive iterations.

FIMIX-PLS returns the segment membership probabilities P_{ik} , which serve as input to compute the relative segment sizes:

$$\rho_k = \frac{\sum_{i=1}^N P_{ik}}{N} \quad (5)$$

Analogously, segment-specific parameter estimates, such as indicator loadings, indicator weights, and path coefficients, are inferred from running weighted regressions using P_{ik} as object weights.

3. Review of FIMIX-PLS research

The research data for our review of FIMIX-PLS use were obtained from the EBSCO Discovery Service online platform. Our search query identifies studies which appeared since the publication of the introductory article on FIMIX-PLS by Hahn, Johnson, Herrmann, and Huber (2002) and that use the following terms in their title, abstract, keywords, or full text: "FIMIX" or "finite mixture", and "PLS" or "partial least squares". Our review of FIMIX-PLS research covers all journals that Scopus indexed in business-related categories (i.e., *Business, Management and Accounting; Business and International Management; Marketing; Strategy and Management; Management Science and Operations Research; Organizational Behavior and Human Resource Management*, and *Tourism, Leisure and Hospitality Management*), ranked in the first or second quartile (i.e., Q1 and Q2).

Table 1
Studies using FIMIX-PLS.

Journal	Author(s)
<i>Annals of Tourism Research</i>	Marques and Reis (2015)
<i>Asia Pacific Journal of Marketing & Logistics</i>	Le, Tran, Nguyen, and Cheng (2019)
<i>Engineering Economics</i>	Alonso-Dos-Santos, Vveinhardt, Calabuig-Moreno, and Montoro-Rios (2016)
<i>European Business Review</i>	Matthews, Sarstedt, Hair, and Ringle (2016)
<i>European Journal of Marketing</i>	Carlson, Gudergan, Gelhard, and Rahman (2019)
	Dessart, Aldás-Manzano, and Veloutsou (2019)
	Piehlér, King, Burmann, and Xiong (2016)
	Willems, Brengman, Van Kerrebroeck (2019)
<i>Group Decision and Negotiation</i>	Niemeyer, Teubner, Hall, and Weinhardt (2018)
<i>Industrial Management & Data Systems</i>	Alonso Dos Santos, Rejón Guardia, and Calabuig Moreno (2018)
	Gong, Zhang, Chen, Zhao, and Lee (2019)
	Inigo, Ritala, and Albareda (2020)
<i>Industrial Marketing Management</i>	Kamath, Pai, and Prabhu (2020)
<i>International Journal of Bank Marketing</i>	Thamaraiselvan, Sridevi, Senthil Arasu, and Srinivasan (2018)
<i>International Journal of Business Excellence</i>	Sánchez-Fernández, Jiménez-Castillo, and Iniesta-Bonillo (2017)
<i>International Journal of Market Research</i>	Latan, Chiappetta Jabbour, Lopes de Sousa Jabbour, de Camargo Fiorini, and Foropon (2020)
<i>International Journal of Production Economics</i>	Teller and Gittenberger (2011)
<i>International Review of Retail, Distribution & Consumer Research</i>	Robinson and Kalafatis (2020)
<i>Journal of Advertising Research</i>	Abu Farha and Elbanna (2018)
<i>Journal of Business & Industrial Marketing</i>	Silva, Pinho, Soares, and Sá (2019)
<i>Journal of Business Economics and Management</i>	Barhorst, Wilson, and Brooks (2020)
<i>Journal of Business Research</i>	Gelhard, von Delft, and Gudergan (2016)
	Valette-Florence, Guizani, and Merunka (2011)
	Mondéjar-Jiménez, Segarra-Oña, Peiró-Signes, Payá-Martínez, and Sáez-Martínez (2015)
<i>Journal of Cleaner Production</i>	Londono, Wilson, and Osorio-Tinoco (2020)
<i>Journal of Entrepreneurship in Emerging Economies</i>	Abu Farha, Al-Kwafi, and Ahmed (2018)
<i>Journal of Hospitality & Tourism Technology</i>	Zhang, Wang, and Hao (2020)
<i>Journal of Knowledge Management</i>	Janka and Guenther (2018)
<i>Journal of Management Accounting Research</i>	Ahrholdt (2011)
<i>Journal of Marketing Theory & Practice</i>	Navarro, Acedo, Losada, and Ruzo (2011)
	Huang (2020)
<i>Journal of Organizational and End User Computing</i>	Arenas-Gaitán, Rondan-Cataluña, and Ramírez-Correa (2018)
<i>Journal of Research in Interactive Marketing</i>	Chocarro, Cortiñas, and Villanueva (2015)
	Groß (2018)
<i>Journal of Retailing and Consumer Services</i>	Matzler, Strobl, Thurner, and Füller (2015)
<i>Journal of Service Management</i>	Schirmer, Ringle, Gudergan, and Feistel (2018)
<i>Journal of Strategic Marketing</i>	Wilden and Gudergan (2015)
<i>Journal of the Academy of Marketing Science</i>	Zhou, Fey, and Yildiz (2020)
<i>Journal of World Business</i>	Klarner, Sarstedt, Hoeck, and Ringle (2013)
<i>Long Range Planning</i>	Money, Hillenbrand, Henseler, and da Camara (2012)
	Stewart Jr., May, and Ledgerwood (2015)
<i>Management International Review</i>	Batouei, Iranmanesh, Mustafa, Nikbin, and Ping (2020)
<i>Research in Transportation Business and Management</i>	Malca, Peña-Vinces, and Acedo (2020)
<i>Small Business Economics</i>	Sarstedt, Ringle, Cheah, Ting, Moisescu, and Radomir (2020)
<i>Tourism Economics</i>	Papastathopoulos, Kaminakis, and Mertzanis (2020)
<i>Tourism Management Perspectives</i>	

Our search initially retrieved 137 studies. We then coded all these studies independently to identify FIMIX-PLS applications. Ultimately, we excluded 92 studies from the analysis, as they did not apply the method, but referred to FIMIX-PLS-related articles to support another aspect of their study. It is worth noting that 18 of these 92 studies emphasize the importance of conducting a FIMIX-PLS analysis, and recommend that researchers should apply it in future to either demonstrate their model's stability, or to uncover and treat unobserved heterogeneity. The final list comprises 45 studies, most of which were published in Q1 journals (32 studies; 71.11%)—see Table 1. The majority of journals only published a single FIMIX-PLS study, the exceptions being the *European Journal of Marketing* (4 studies), *Journal of Business Research* (3 studies), *Long Range Planning*, *Journal of Marketing Theory & Practice*, *Industrial Management & Data Systems*, and *Journal of Research in Interactive Marketing* (2 studies each). The 45 studies were published over a ten-year period between 2011 and 2020, with more than half (28 of the 45 studies) published in the past three years, revealing the method's increasing relevance for PLS-SEM research. The growing number of papers applying FIMIX-PLS could potentially be attributed to guideline articles (Hair, Sarstedt, Matthews, & Ringle, 2016; Matthews, Sarstedt, Hair, & Ringle, 2016) introducing the method to a broader audience.

4. Critical issues in FIMIX-PLS applications

Each study was evaluated according to a wide range of criteria in order to identify critical issues and potential misapplications of the FIMIX-PLS method. Our review focuses on the following key issues in the application of FIMIX-PLS: (1) determining the number of segments by using information and classification criteria, (2) the segment sizes, (3) segment-specific path coefficient estimates, (4) follow-up latent class analyses, and (5) the ex post analysis.

4.1. Determining the number of segments by using information and classification criteria

The key advantage of FIMIX-PLS over alternative PLS-SEM-based latent class analysis techniques is that the method offers guidance on how many segments to retain from the data. Specifically, the log-likelihood formulation in Eq. (4) facilitates the computation of information criteria, which are well known from the regression literature (Andrews & Currim, 2003; Becker, Ringle, Sarstedt, & Völckner, 2015; Sawa, 1978; Vrieze, 2012). Information criteria simultaneously take a model's fit (i.e., the likelihood value) and the number of parameters used to achieve this fit into account by complementing the resulting (negative) likelihood value with a penalty term, which increases with the number of segments (Burnham & Anderson, 2002). Smaller criterion values indicate a better-fitting segmentation solution. Sarstedt, Becker, Ringle, and Schwaiger (2011) have evaluated different information criteria's efficacy across a broad range of data and model constellations in FIMIX-PLS. Their results demonstrate that researchers should jointly consider Akaike Information Criterion with factor 3 (AIC₃; Akaike, 1973) and Consistent AIC (CAIC; Bozdogan, 1987). Whenever these two criteria indicate the same number of segments, this solution is likely the most appropriate. AIC with factor 4 (AIC₄; Bozdogan, 1994) and BIC generally perform well, while other criteria exhibit pronounced under or overestimation tendencies. The latter specifically holds for AIC, which often overspecifies the correct number of segments, whereas Minimum Description Length 5 (MDL₅; Liang, Jaszak, & Coleman, 1992) shows pronounced underestimation tendencies. Researchers can use this information to determine a certain range of reasonable segment numbers. See Table A1 in the Appendix for a formal presentation of the above criteria.

The decision regarding how many segments to retain should also consider classification criteria, which express the segments' degree of fuzziness (Wedel & Kamakura, 2000). While authors have proposed several classification criteria (see Chaps. 6.8–6.10 in McLachlan & Peel, 2000 for an overview), the only criterion discussed in the context of FIMIX-PLS is Ramaswamy, DeSarbo, Reibstein, and Robinson (1993) entropy (EN) statistic, defined as

$$EN = 1 - \frac{(\sum_{i=1}^N \sum_{k=1}^K - P_{ik} \ln P_{ik})}{N \ln K}$$
 (6)

The EN takes values between 0 and 1, with lower values indicating a higher degree of fuzziness. Researchers generally assume that an EN value of 0.5 and higher reflects sufficient consistency. Conversely, extremely high EN values of 0.90 and above indicate that the solutions overfit the data. Such a situation could occur if there are many missing values in the data or if researchers apply a mean value replacement, which maximizes homogeneity in objects with imputed case values (Hair, Sarstedt, Matthews, & Ringle, 2016). While the EN is not a suitable stand-alone criterion for determining the number of segments

(Sarstedt, Becker, Ringle, & Schwaiger, 2011), it ensures that the information criteria's indicated segmentation solution is not subject to considerable fuzziness that would attenuate the results' generalizability (Ramaswamy, DeSarbo, Reibstein, & Robinson, 1993).

Our analysis shows that in 32 of 45 studies (71.11%), the authors report and rely on information criteria to decide whether unobserved heterogeneity poses a problem and—if applicable—to decide on a suitable number of segments to retain from the data (Table 2). Conversely, in 4 of the 45 studies, the authors draw on information criteria to choose between different segmentation solutions, but fail to report the relevant values. The remaining nine studies either report information criteria, but fail to build on the results (1 study), or do not report or use criteria values at all (8 studies).

Surprisingly, 3 of the 45 studies rely solely on segment sizes and the EN criterion to determine the number of segments. Given the EN criterion's poor performance in terms of selecting an appropriate number of segments (Sarstedt, Becker, Ringle, & Schwaiger, 2011) and the ambiguities that only focusing on segment sizes causes (Rigdon, Ringle, & Sarstedt, 2010), this practice is problematic. Fortunately, only a minority of studies is subject to potentially invalid results by solely assessing the EN criterion.

Table 3 provides a breakdown of the criteria used to determine the number of segments in the above-mentioned 32 studies. We find that most studies rely on the CAIC (28 studies) and BIC (23 studies), followed by the AIC (20 studies) and AIC₃ (17 studies). Only 17 of the 32 studies (53.13%) follow the best practice by considering the AIC₃ and CAIC simultaneously.

Analyzing the number of criteria considered, we find that the studies draw on an average (median) of three criteria and a maximum of seven (Dessart, Aldás-Manzano, & Veloutsou, 2019). Four studies rely on a single information criterion, one of which is the AIC, which tends to overestimate the number of segments in FIMIX-PLS (Sarstedt, Becker, Ringle, & Schwaiger, 2011). In several of the studies (12 of the 32; 37.5%), the authors report more information criteria than they actually considered when determining the number of segments. Conversely, the authors of one study base their decision on the number of segments on four information criteria, but only report two of them.

Overall, the authors' application of information criteria leaves a lot to be desired. While most of the studies draw on information criteria when deciding on the number of segments, several studies do not, despite their fundamental importance for PLS-SEM-based latent class analysis. While many studies rely on appropriate criteria for their analysis (AIC₃ and CAIC), others strongly or solely rely on criteria such as the AIC, MDL₅, and other metrics (EN and relative segment sizes), which are inappropriate in this context. In addition, several authors fail to adequately report the criteria considered in their analysis, making it difficult to assess the segment selection's validity.

In terms of entropy, we find that 25 studies consider and report the EN statistic in their analyses—the other studies either make no reference to the EN (14 of the 45 studies; 31.11%), consider the metric, but do not report its values (2 studies; 4.44%), or refer to the EN statistic, although they consider a one-segment solution in the final analysis (4 studies; 8.88%). A detailed analysis of the 25 studies that consider and report the EN criterion gives rise to concern. While the average EN value of 0.65

Table 2
Reporting and use of information criteria.

Information criteria	Used	Not used
Reported	32 (71.11%)	1 (2.22%)
Not reported	4 (8.89%)	8 (17.78%)

Table 3
Use of information criteria.

Criterion	Number of studies reporting (n = 32)
Consistent AIC (CAIC)	28 (87.50%)
Bayesian Information Criterion (BIC)	23 (71.88%)
Akaike Information Criterion (AIC)	20 (62.50%)
Modified AIC 3 (AIC ₃)	17 (53.13%)
Modified AIC 4 (AIC ₄)	7 (21.88%)
Minimum Description Length 5 (MDL ₅)	8 (25.00%)
Hannan-Quinn information criterion (HQ)	1 (3.13%)

(median = 0.62) ensures that the segmentation solution is sufficiently consistent, 5 of the 25 studies (20%) draw on a segmentation solution with an EN value of below 0.5. Research findings with such low EN values should be considered with caution, as the results suggest that the objects' partitioning does not reflect the data heterogeneity adequately, which calls for an aggregate data level analysis (Sarstedt et al., 2017). However, five studies report EN values of 0.90 and higher, indicating that the segmentation solution potentially overfits, which has adverse consequences for the results' generalizability.

4.2. Segment sizes

Segment size—the number of objects assigned to each segment—is another important aspect of any FIMIX-PLS application. FIMIX-PLS reports the segment sizes that it infers from the probabilities of the objects' segment membership (see Eq. (5)), which results in overlapping segments. Researchers therefore need to engage in a hard clustering by assigning each object to the segment with the highest membership probability; that is, $\max(P_{ik})$ (Nagin, 2005). In doing so, researchers should ensure that their segments include a sufficient number of observations to reliably estimate the model parameters. This step is of fundamental importance in FIMIX-PLS, as the EM algorithm always converges to the prespecified number of segments K (Hahn, Johnson, Herrmann, & Huber, 2002), which can, however, produce satellite segments that do not meet the minimum sample size requirements.

Our analysis shows that 34 of the 45 studies employing FIMIX-PLS extracted multiple segments from the data. Specifically, 17 studies (50%) considered 2 segments, 11 studies (32%) extracted 3 segments,

while the other 6 studies (18%) considered 4 segments (Table 4). In the remaining 11 studies, the authors concluded that a significant degree of heterogeneity did not impact their results, because, for example, the information criteria converged to a one-segment solution or because higher-segment solutions were too fuzzy. The majority of studies that extracted multiple segments (23 of the 34 studies; 67.65%) reported segment sizes after a hard clustering of the objects, using the segment membership probabilities as input (Table 4). We identified one study which failed to report any segment size, and another two which only reported segment sizes resulting from the FIMIX-PLS output. This practice is problematic, since the relative segment sizes from the FIMIX-PLS output are only inferred from the segment membership probabilities through weighted regressions (Sarstedt, Ringle, & Gudergan, 2016).

Table 4 also reports the smallest segment size analyzed in each study after a hard clustering of the objects. We find considerable differences across the studies, with segment sizes ranging between 21 and 259 (mean = 97) objects in the smallest segment. We also examine whether the 24 studies that hard cluster the objects on the basis of FIMIX-PLS probabilities comply with the ten times rule concerning the minimum sample size required for PLS-SEM (Hair et al., 2022), although this size has been identified as very liberal (Goodhue, Lewis, & Thompson, 2012; Kock & Hadaya, 2018; Rigdon, 2016). In two studies (Arenas-Gaitán, Rondon-Cataluña, & Ramírez-Correa, 2018; Le, Tran, Nguyen, & Cheng, 2019), the smallest segments didn't meet the ten times rule. However, the researchers dropped the corresponding segments and focused the interpretation of their results on the larger segments.

In the next step, we analyzed the statistical power of the studies' segment-specific analyses on the basis of Kock and Hadaya (2018)

Table 4
Relative segment sizes in studies resulting in multiple segment solutions.

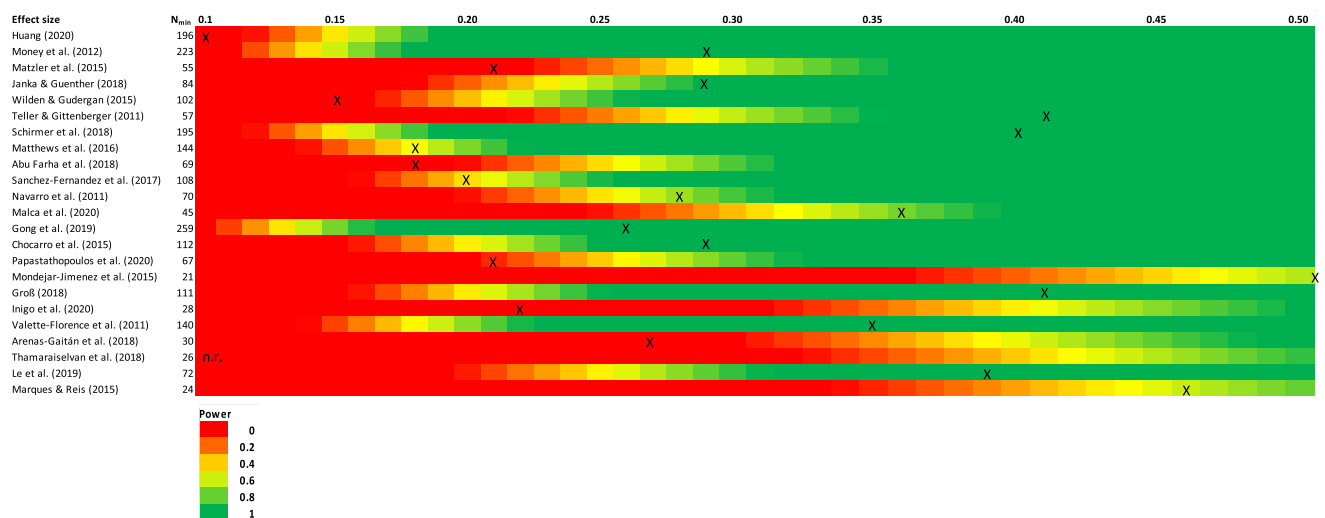
No	Study	Total Sample size	No. of segments	FIMIX-PLS output				FIMIX-PLS hard clustering				$N_{\min}$
				k = 1	k = 2	k = 3	k = 4	k = 1	k = 2	k = 3	k = 4	
1	Huang (2020)	627	2	0.715	0.285			0.687	0.313			196
2	Money et al. (2012)	501	2	0.593	0.407			0.555	0.445			223
3	Matzler et al. (2015)	327	2	0.817	0.183			0.832	0.168			55
4	Janka and Guenther (2018)	276	2	0.638	0.362			0.696	0.304			84
5	Wilden and Gudergan (2015)	228	2	0.509	0.491			0.553	0.447			102
6	Teller and Gittenberger (2011)	404	2	–				0.859	0.141			57
7	Schirmer et al. (2018)	749	2	–				0.740	0.260			195
8	Matthews et al. (2016)	336	2	–				0.571	0.429			144
9	Abu Farha et al. (2018)	210	2	–				0.671	0.329			69
10	Sanchez-Fernandez et al. (2017)	500	2	–				0.784	0.216			108
11	Navarro et al. (2011)	150	2	–				0.533	0.467			70
12	Malca et al. (2020)	95	2	–				0.526	0.474			45
13	Gong et al. (2019)	552	2	–				0.531	0.469			259
14	Londono et al. (2020)	643	2	0.784	0.216			–				–
15	Silva et al. (2019)	524	2	0.696	0.304			–				–
16	Alonso Dos Santos et al. (2016)	259	2	–				–				–
17	Alonso Dos Santos et al. (2018)	609	2	0.864	0.136			–				–
18	Chocarro et al. (2015)	972	3	–				0.580	0.300	0.120		112
19	Papastathopoulos et al. (2020)	559	3	–				0.576	0.304	0.120		67
20	Mondejar-Jimenez et al. (2015)	223	3	–				0.812	0.094	0.094		21
21	Groß (2018)	734	3	–				0.619	0.230	0.151		111
22	Inigo et al. (2020)	167	3	–				0.569	0.263	0.168		28
23	Valette-Florence et al. (2011)	538	3	–				0.409	0.331	0.260		140
24	Abu Farha & Elbanna (2018)	245	3	–				–				–
25	Niemeyer et al. (2018)	216	3	0.500	0.264	0.236		–				–
26	Dessart et al. (2019)	970	3	0.684	0.201	0.115		–				–
27	Robinson and Kalafatis (2020)	315	3	0.476	0.314	0.210		–				–
28	Carlson et al. (2019)	782	3	0.600	0.390	0.010		–				–
29	Arenas-Gaitán et al. (2018)	868	4	–				0.740	0.214	0.035	0.012	30 [†]
30	Thamaraivelan et al. (2018)	294	4	–				0.357	0.354	0.201	0.088	26
31	Le et al. (2019)	403	4	–				0.397	0.365	0.179	0.060	72 [†]
32	Marques & Reis (2015)	381	4	–				0.262	0.381	0.294	0.063	24
33	Kamath et al. (2020)	500	4	0.450	0.202	0.186	0.162	–				–
34	Gelhard et al. (2016)	99	4	0.465	0.212	0.182	0.141	–				–

Notes: [†] The authors only analyzed the three largest segments; $N_{\min}$ indicates the size of the smallest analyzed segment after hard clustering.

Table 5

The application of FIMIX-PLS: Issues and recommendations.

Issue	Recommendation	References
Determining the number of segments	Consider AIC ₃ and CAIC jointly. Use criteria with a strong tendency to underestimate (MDL ₅) and overestimate (AIC) the number of segments in order to define a reasonable range of segments. Select a number of segments with a sufficient degree of consistency (EN ≥ 0.5).	Sarstedt et al. (2011)
Segment sizes	Ensure that the segments meet the minimum sample size requirements. Consider only segments that meet the minimum sample size requirements.	Kock and Hadaya (2018)
Segment-specific parameter estimates	Do not interpret path coefficient estimates that FIMIX-PLS provides. Use estimates from a hard clustering and, specifically, from a partitioning after undertaking an ex post analysis.	Rigdon et al. (2010); Sarstedt et al. (2016b)
Follow-up latent class analysis	Use the FIMIX-PLS result as the starting partitioning of a follow-up latent class analysis by using PLS-POS, PLS-GAS, or PLS-IRRS.	Becker et al. (2013), Sarstedt et al. (2017)
Ex post analysis	Identify explanatory variables that yield a similar partitioning of objects as FIMIX-PLS after a hard clustering. The overlap should be at least 60%.	Ringle et al. (2011)



Notes: X=median effect size considered in each study's analysis of the smallest segment. n.r.=not reported

Fig. 1. Heat map of statistical power of different effect sizes. Notes: X = median effect size considered in each study's analysis of the smallest segment. n.r. = not reported.

inverse square root method for effect sizes between 0.1 and 0.5, and assuming a significance level of 5%. The heat map in Fig. 1 shows that only around half of the studies achieve power levels of 80% when taking the median effect sizes in each study as the standard of comparison. Nine of the 23 studies (39.12%) are severely underpowered, with the smallest segments' analysis yielding power levels of less than 50% in terms of the relevant median effect. In many cases, the segment-specific sample sizes do not even allow for disclosing moderate to high effects of 0.30 and above. These results point to severe issues with the application of FIMIX-PLS, because segment-specific analyses need to build on sufficient sample sizes to warrant reliable and valid findings.

4.3. Segment-specific path coefficient estimates

FIMIX-PLS derives segment-specific path coefficients by running weighted regressions of each endogenous construct on its direct predecessor constructs in the structural model, using P_{ik} (Eq. (5)) as the object weights. As such, the path coefficient estimates fail to offer concrete insights into hypothesis testing or into deriving concrete managerial recommendations (Sarstedt, Ringle, & Gudergan, 2016). Researchers should therefore report the path coefficients when estimating each segment derived from a hard clustering of objects (Clogg, 1995).

Our review documents that only two of the 34 studies (5.88%) with extracted multiple segments report the FIMIX-PLS path coefficients. In fact, in these two studies, the authors only report and interpret the abstract path coefficients that the FIMIX-PLS analysis produced, and do not perform a hard clustering. Conversely, with the exception of one study, in all the studies that engage in a hard clustering of the objects (i.e., 23 of the 24 studies; 95.83%), researchers report the segment-specific path coefficients after assigning the objects to these segments on the basis of their highest probability of segment membership. Our review shows a similar pattern of researchers reporting their models' explanatory power on the basis of the R^2 statistic (Hair, Risher, Sarstedt, & Ringle, 2019). While one study reports the R^2 values as derived from the weighted regressions, which offer very limited insight into the model's explanatory power, 20 studies (83.33%) provide this information on the basis of the hard clustering of objects.

4.4. Follow-up latent class analysis

While FIMIX-PLS proves valuable for assessing whether or not a significant degree of heterogeneity impacts the aggregate results, the method does not generally reproduce the segment-specific effects well (Becker, Rai, Ringle, & Völckner, 2013; Ringle et al., 2009; Ringle,

Sarstedt, & Schlittgen, 2014). Specifically, FIMIX-PLS usually overfits one segment, thereby producing an extremely high R^2 value, while the other segments are treated as residual ones with low explanatory power. Researchers address this concern by proposing alternative latent class methods in a PLS-SEM framework.

Ringle, Sarstedt, Schlittgen, and Taylor (2013), for example, introduced the PLS genetic algorithm segmentation (PLS-GAS) approach, which relies on a search heuristic that gradually improves the segmentation solution in a procedure akin to an evolutionary process. The method uses guided random searches to reveal a local optimum solution, which a deterministic hill-climbing approach then gradually improves. PLS-GAS performs very well in terms of disclosing the actual segment structure, but is computationally very demanding, which has led to the development of the PLS iterative reweighted regressions (PLS-IRRS) approach.

Drawing on a robust regression framework (Schlittgen, 2011), PLS-IRRS identifies objects with extreme case values, which are regarded as separate segments. Schlittgen, Ringle, Sarstedt, and Becker (2016) simulation study shows that its results match those of PLS-GAS, but with considerably lower computational power demands.

Finally, Becker, Rai, Ringle, and Völckner (2013) introduced the PLS prediction-oriented segmentation, a hill-climbing approach that gradually reallocates objects between the segments in order to maximize the solution's (weighted) R^2 . The authors' simulation study shows that PLS-POS performs segmentation well, providing favorable outcomes compared with alternative latent class techniques.

Owing to its performance and implementation in popular PLS-SEM software, such as SmartPLS (Ringle, Wende, & Becker, 2015; Sarstedt & Cheah, 2019), Sarstedt et al. (2017) recommend the tandem use of FIMIX-PLS and PLS-POS. Researchers should therefore use FIMIX-PLS to assess whether or not heterogeneity impacts the aggregate level analysis, and—if necessary—should identify a suitable number of segments. The partitioning of objects based on FIMIX-PLS after a hard clustering should subsequently serve as input for a follow-up PLS-POS latent class analysis.

Our analysis shows that only few studies follow this approach. In fact, only 4 of the 34 studies (11.76%) that identify multiple segments also apply another latent class technique. Specifically, Alonso Dos Santos et al. (2018), Dessart, Aldás-Manzano, and Veloutsou (2019), Kamath, Pai, and Prabhu (2020), and Robinson and Kalafatis (2020) employ PLS-POS after initially using FIMIX-PLS to establish the appropriate number segments. Klarner, Sarstedt, Hoeck, and Ringle (2013) also use PLS-GAS, but only to confirm a one-segment solution.

These results are cause for concern, since the original FIMIX-PLS results do not mirror the actual segment-specific effects well. Researchers should be cautious when following recommendations for follow-up analyses if they wish to achieve adequate results.

4.5. Ex post analysis

While the results from the hard clustering of objects offers actionable insights beyond the original FIMIX-PLS, the resulting segments are—by definition—latent. Researchers have therefore called for the partitioning of the data by means of an explanatory variable, or a combination of several, to produce a grouping of data that largely corresponds to the one that FIMIX-PLS indicates after hard clustering (Hair, Sarstedt, Matthews, & Ringle, 2016; Rigdon, Ringle, & Sarstedt, 2010; Sarstedt & Ringle, 2010). Performing such an ex post analysis is mandatory—otherwise the results remain abstract and have no reference to any observable characteristics (Sarstedt, Ringle, & Gudergan, 2016). Nonetheless, Hair, Sarstedt, Matthews, and Ringle (2016) note that “reproducing the FIMIX-PLS partition remains a very challenging task, as observable characteristics often do not match the latent segment structures well. Against this background, an overlap of 60 per cent between the FIMIX-PLS partition and the one produced by the explanatory variable(s) can be considered satisfactory.” Suitable methods for disclosing

relevant explanatory variables include crosstabs, classification and regression trees, and logistic regression (Ringle, Sarstedt, & Mooi, 2010).

We find that only 14 of the 34 studies (41.17%) that extract multiple segments perform an ex post analysis. However, in six of these studies, the authors do not try to reproduce the partition by hard clustering the objects, but use the explanatory variables to characterize the segments. In two other studies, the authors indicate that they are unable to identify satisfactory explanatory variables (Le, Tran, Nguyen, & Cheng, 2019; Matzler, Strobl, Thurner, & Füller, 2015). This finding underlines the importance of considering as many explanatory variables in the data collection as possible in order to provide a sufficient basis for a successful ex post analysis. The remaining six studies follow best practice by using explanatory variables to reproduce the partition by hard clustering the objects as closely as possible. However, only four of these studies also report the degree of segment overlap with the original partition. In addition, these studies generally indicate a satisfactory overlap of between 59.8% (Matthews, Sarstedt, Hair, & Ringle, 2016) and 93.7% (Huang, 2020), with an average value of 71.77%. The other two studies do not reveal the overlap, which makes it difficult to assess the explanatory variables' suitability for capturing the heterogeneity that the FIMIX-PLS analysis identifies.

Finally, we analyze the types of explanatory variables used to characterize or reproduce the latent segments. Commonly used explanatory variables refer to demographics (6 studies: Chocarro, Cortiñas, & Villanueva, 2015; Huang, 2020; Money, Hillenbrand, Henseler, & Da Camara, 2012; Papastathopoulos, Kaminakis, & Mertzanis, 2020; Schirmer, Ringle, Gudergan, & Feistel, 2018; Thamaraiselvan, Srivevi, Arasu, & Srinivasan, 2018), behaviors (Money, Hillenbrand, Henseler, & Da Camara, 2012; Teller & Gittenberger, 2011), perceptions (Arenas-Gaitán, Rondan-Cataluña, & Ramírez-Correa, 2018), company characteristics (Abu Farha, Al-Kwif, & Ahmed, 2018; Janka & Guenther, 2018), service types (Matthews, Sarstedt, Hair, & Ringle, 2016), and the competitive environment (Wilden & Gudergan, 2015).

To summarize, many researchers working with FIMIX-PLS do not undertake an ex post analysis. Those that do, document its results haphazardly, making it difficult to appreciate this central aspect of the latent class analysis.

5. Discussion and conclusion

With PLS-SEM's increasing popularity in business research (Ghasemy, Teeroovengadum, Becker, & Ringle, 2020; Lin et al., 2020; Usakli & Kucukergin, 2018), the use of latent class analysis techniques should give the method further momentum. Latent class analyses have two aims: First, to identify and—if necessary—treat unobserved heterogeneity, and, second, to ensure that the data patterns' analysis does not substantially impact the aggregate data. Given the substantial biases that heterogeneity causes, Sarstedt and Ringle (2010, 1316) note that researchers should routinely apply the FIMIX-PLS approach to latent class analysis in order to “decide whether or not unobserved heterogeneity affects path modeling outcomes.” Numerous research groups have echoed this call over the last few years (e.g., Latan, 2018; Matthews, Hair, & Matthews, 2018; Russo & Stol, 2021). Tying in with this research strand, we critically review FIMIX-PLS use in business research-related journals with the aim of establishing a foundation for the method's future applications. Our review identifies several areas for improvement—based on our observations, we offer a series of recommendations that we summarize in Table 5.

First, researchers' use and reporting of information criteria require significant improvement. As evidenced in our review, authors often fail to consider information and classification criteria when choosing the final segmentation solution, although this is of fundamental importance for PLS-SEM-based latent class analysis. In addition, many authors fail to adequately report the criteria they consider in their analysis, which complicates assessing how valid the segment solution selection is. In this

respect, we strongly recommend that researchers routinely use and report information criteria—at least in an appendix. When doing so, researchers should be aware that some criteria have a strong tendency to overspecify (AIC) or underspecify (MDL_5) the correct number of segments (Sarstedt, Becker, Ringle, & Schwaiger, 2011). Choosing an appropriate number of segments is generally not a binary task, but requires researchers to identify a suitable range of segments on the basis of information and classification criteria. The subsequent decision regarding the number of segments to retain and to potentially stick to a one-segment solution should also consider other elements, such as the EN statistic and relative segment sizes (Sarstedt et al., 2017). Our review shows that very few researchers follow this approach.

Second, our analysis reveals that many researchers do not engage in a hard clustering of segments on the basis of the segment membership probabilities that FIMIX-PLS produces. Those who do, often rely on partitions with small segments, which produce very modest levels of statistical power. We strongly recommend researchers to pay greater attention to the segment sizes in order to ensure that group-specific model estimations achieve power levels of at least 80%. While the ten times rule does offer initial guidance (Hair, Sarstedt, & Ringle, 2019), researchers should primarily rely on more elaborate procedures to determine the minimum sample sizes (e.g., Aguirre-Urreta & Rönkkö, 2015; Kock & Hadaya, 2018). Such an analysis might indicate that researchers should extract fewer segments than the information and classification criteria indicate. Researchers could, however, proceed with the analysis, but focus their interpretation on those segments that rely on sufficient sample sizes.

Third, although the path coefficient estimates that FIMIX-PLS produces are abstract and do not offer any grounds for hypothesis testing or for deriving concrete managerial recommendations, various researchers do document and interpret these coefficients. Future studies should avoid this *modus operandi* and, instead, provide segment-specific model estimations based on the hard clustering of objects.

Fourth, even though undertaking an ex post analysis of the segments based on FIMIX-PLS probabilities is mandatory in order to translate the results into observable segments, the vast majority of the studies in our review does not engage in such an analysis. The few researchers that run an ex post analysis do not consistently report the degree of segment overlap with the hard clustering partition. Such reporting is, however, required to confirm the explanatory variables' adequacy regarding capturing the heterogeneity that the FIMIX-PLS analysis identifies. Researchers should therefore put far more emphasis on translating the latent segments that FIMIX-PLS identifies into observable segments by using an ex post analysis. Furthermore, researchers should facilitate this analysis by already planning the ex post analysis in the data collection stage and including as many potentially explanatory variables as possible.

Our findings also suggest several avenues for future research. For example, researchers seem to struggle to identify suitable explanatory variables in their ex post analysis. While the studies in our review rely on a wide range of methods, such as crosstabs (e.g., Arenas-Gaitán, Rondan-Cataluña, & Ramírez-Correa, 2018), classification and regression trees (e.g., Alonso Dos Santos et al., 2018), logistic regressions (e.g., Dessart, Aldás-Manzano, & Veloutsou, 2019), and analyses of variance (e.g.,

Robinson & Kalafatis, 2020), research still lacks a procedure to systematically match the latent segmentation that FIMIX-PLS produces with observable characteristics. Similarly, the notion of overfit, which excessive EN values indicate, needs further elaboration. The missing value treatment's role and impact on segment retention in PLS-SEM-based latent class analyses have not at yet been explored. In addition, researchers need to explore other metrics to quantify the degree of fuzziness, such as—in an overfit context—Bezdek's (1981) partition coefficient and partition entropy.

Future research should also extend the FIMIX-PLS algorithm by introducing constraints to ensure that the extracted segments meet the minimum standards in terms of the sample sizes. Such constraints could introduce a penalty function that, depending on the model complexity, changes the objects' segment membership probabilities in order to fit them into K segments, with each segment meeting the minimum sample size. Research on factor-based SEM suggests that partitioning objects on the basis of their segment membership probabilities attenuates the model estimates (Bolck, Croon, & Hagenars, 2004; Vermunt, 2010). To address this concern, researchers have proposed including explanatory variables as covariates in the model estimation (Bray, Lanza, & Tan, 2015). Future studies could implement a similar procedure in PLS-SEM-based latent class analysis. FIMIX-PLS applications could similarly implement multiple pseudo-class draws (Bandeem-Roche, Miglioretti, Zeger, & Rathouz, 1997) to translate the latent segmentation into an observed one. Using this approach, objects are multiple times and randomly classified into latent classes based on the distributions (rather than the maximum) of their segment membership probabilities. Unlike the hard clustering approach commonly used in FIMIX-PLS, and which guidelines advocate (Hair et al., 2016; Hair, Sarstedt, Ringle, & Gudergan, 2018; Rigdon, Ringle, & Sarstedt, 2010), this approach explains uncertainty in the segment assignment. Future research should therefore evaluate this approach's efficacy in a FIMIX-PLS context.

CRedit authorship contribution statement

Marko Sarstedt: Conceptualization, Methodology, Formal analysis, Writing – original draft, Writing – review & editing, Visualization, Supervision. **Lăcrămioara Radomir:** Conceptualization, Investigation, Data curation, Writing – original draft, Writing – review & editing. **Ovidiu Ioan Moisesescu:** Conceptualization, Investigation, Data curation, Writing – original draft, Writing – review & editing. **Christian M. Ringle:** Conceptualization, Methodology, Writing – original draft, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A

Table A1
Information criteria.

Criterion	Description	Reference
Akaike Information Criterion	$AIC = -2 \cdot \ln L + 2 \cdot N_k$	Akaike (1973)
Modified AIC 3	$AIC_3 = -2 \cdot \ln L + 3 \cdot N_k$	Bozdogan (1994)
Modified AIC 4	$AIC_4 = -2 \cdot \ln L + 4 \cdot N_k$	Bozdogan (1994)
Consistent AIC	$CAIC = -2 \cdot \ln L + [\ln(N) + 1] \cdot N_k$	Bozdogan (1987)
Bayesian Information Criterion	$BIC = -2 \cdot \ln L + \ln(N) \cdot N_k$	Schwarz (1978)
Minimum Description Length 5	$MDL_5 = -2 \cdot \ln L + 5 \cdot \ln(N) \cdot N_k$	Liang et al. (1992)

Notes: $\ln L$ = log-likelihood; N_k = the number of parameters required to estimate the model with k segments; N = sample size.

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